

A Fixed-Direction Ephemeris Test of $\bar{s}^{\text{TX}}$ at the CMB Apex: Pre-Registered Prediction, Analysis Recipe, and a Provisional Bound from Public Data

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Abstract

Planetary-ephemeris constraints on the boost sector of the gravitational Standard-Model Extension (SME) are limited not by data quality but by degeneracy: the standard analysis fits the components of $\bar{s}^{\text{TJ}}$ jointly with free orientation, and the published marginal uncertainty on $\bar{s}^{\text{TX}}$ is $\sim 8 \times 10^{-9}$ with correlation coefficients near unity (Hees et al. 2015). A specific physical hypothesis removes the degeneracy. If the MOND acceleration scale reflects a preferred frame — the frame in which the cosmic microwave background is isotropic — then the induced boost vector is **fixed in direction at the CMB dipole apex**, $(l, b) = (264.02^\circ, 48.25^\circ)$, i.e. $(\text{RA}, \text{Dec}) = (167.94^\circ, -6.94^\circ)$ in the Sun-centered frame, with component ratios $\bar{s}^{\text{TY}}/\bar{s}^{\text{TX}} = -0.214$ and $\bar{s}^{\text{TZ}}/\bar{s}^{\text{TX}} = +0.125$ locked. The ephemeris test then collapses to a **one-parameter fit** for the amplitude A , with the de Sitter–Unruh modified-inertia framework predicting $|A| \approx 8.7 \times 10^{-10}$ **with negative $\bar{s}^{\text{TX}}$** — the same sign as the published central value. We supply (i) the pre-registered prediction and its kill condition; (ii) the concrete analysis recipe for an ephemeris group (one added acceleration term, one fitted parameter, no change to light-time modeling); (iii) a *provisional* fixed-direction bound from public secular perihelion/node-advance tables: $A = (-2.5 \pm 5.9) \times 10^{-10}$ (INPOP10a set; $(-0.9 \pm 6.4) \times 10^{-10}$ with INPOP15a updates), i.e. the prediction sits at $\sim 1.4\sigma$ with matching sign — neither detected nor excluded — while the fixed direction alone already sharpens the effective bound to $|A| < 1.2 \times 10^{-9}$ (95%), about twice better than the published marginal; and (iv) an honest scoping discovery: under a *per-body* reading of the underlying relation ($\bar{s} \propto a_0/2g$ per planet) the signal falls 30–1000× below current secular sensitivity, so planetary ephemerides test only the *universal* reading. A dedicated one-parameter fit on archival Mars and MESSENGER ranging reaches an idealized floor $\sigma_A \approx 4 \times 10^{-11}$, with realistic degradation $\sigma_A \sim 1\text{--}4 \times 10^{-10}$: **decidable either way with data already in hand**. The prediction and its kill condition were committed to a public repository before any dedicated fit exists; we accept the kill as written.

1. The degeneracy, and the hypothesis that removes it

The gravitational SME parametrizes leading Lorentz violation in the gravitational sector by a symmetric tensor $\bar{s}^{\mu\nu}$; its boost components $\bar{s}^{\text{TJ}}$ enter planetary equations of motion through a velocity-coupled acceleration (Bailey & Kostelecký 2006). The most complete ephemeris analysis (Hees et al. 2015, INPOP10a secular advances) fits the SME components jointly and reports $\bar{s}^{\text{TX}} = (-2.9 \pm 8.3) \times 10^{-9}$ — consistent with zero, with the large uncertainty driven by near-unit correlations among simultaneously-fitted components, not by ranging noise.

A preferred-frame hypothesis removes the orientation freedom. In a de Sitter–Unruh modified-inertia reading of MOND (Milgrom 1999; the present framework’s realization with $a_0 = c^2\sqrt{(\Lambda/32\pi)} = 9.36 \times 10^{-11} \text{ m/s}^2$), local dynamics carries an imprint of the frame in which the vacuum is isotropic

— observationally, the CMB rest frame. The solar system moves through that frame at $v = 369.82$ km/s toward the dipole apex; the induced boost background is then

$$\bar{s}^{\{TJ\}} = A \cdot \hat{n}_J, \hat{n} \text{ fixed at (RA, Dec)} = (167.94^\circ, -6.94^\circ), \beta = v/c = 1.234 \times 10^{-3},$$

leaving a single amplitude A . The framework’s estimate (committed as `reviews/stx_target.py`; reproduced independently to 1%) is $|A| \approx 8.7 \times 10^{-10}$, **sign negative in $\bar{s}^{\{TX\}}$** . We emphasize scope honestly: the *magnitude* carries the framework’s $O(1)$ modelling freedom; the *direction* and *CPT-even-only* structure are its sharp content, and the CPT-even prediction implies **exactly zero gravitational-wave birefringence** as an independent kill-switch.

2. A load-bearing fork: universal vs per-body reading

Carrying the derivation through the ephemeris observables exposed a fork that any test of this class must state. If the induced $\bar{s}$ is **universal** (one background constant, reading U), planetary secular drifts scale as computed below and the test is sharp. If instead the relation is read **per-body** ($\bar{s} \propto a_0/2g$ evaluated at each planet’s orbital gravity, reading P), the low-gravity outer planets dominate the naive formula but the *observable* drifts fall 30–1000× below current secular uncertainties (best case S/N ≈ 0.3 – 0.4 for Mars); no existing or near-term ranging tests reading P. **All ephemeris statements in this note therefore constrain reading U only**, and the framework’s “in-hand test” claim is hereby scoped to that reading. We found no published statement of this fork; analyses to date implicitly assume U.

3. The recipe (what an ephemeris group runs)

1. Add to each planet’s equation of motion the SME pure-gravity boost term (Bailey & Kostelecký 2006; Hees et al. 2015, Eq. 3): $\delta \ddot{\mathbf{x}} = (2GM_\odot/c^3 r^3)[(\hat{\mathbf{x}}\hat{\mathbf{x}} - \frac{1}{3}\hat{\mathbf{x}}\hat{\mathbf{x}})\hat{\mathbf{x}} - (\hat{\mathbf{x}}\hat{\mathbf{x}})\hat{\mathbf{x}}]$, with $\hat{\mathbf{x}} = A\hat{n}$ and $\hat{n}$ **fixed** as above (component ratios locked; $\bar{s}^{\{TT\}}$ unobservable here; $\bar{s}^{\{jk\}} = 0$).
2. No modification to light-time/Shapiro modeling (boost components are negligible in propagation at this sensitivity).
3. Fit the single parameter A jointly with the standard ephemeris set (initial conditions, $GM_\odot$, asteroid ring). The fixed direction is the entire leverage: it converts an 8-parameter, 0.99-correlated problem into a one-parameter fit $\sim 100\times$ tighter.
4. Report both variants: (U) universal A ; (P) per-body $A(g_{\text{Saturn}}/g_{\text{planet}})$ — the latter as the framework’s own-reading control, expected unconstrained.
5. Detection signature: $A < 0$ ($n_X < 0$), amplitude $\sim 8.7 \times 10^{-10}$, direction consistent with the apex. Kill: $|A| < 2\sigma$ with $\sigma_A \leq 4.3 \times 10^{-10}$ excludes the prediction at $\geq 2\sigma$; the idealized (white-noise) combined floor from archival MESSENGER + Mars ranging is $\sigma_A \approx 4.3 \times 10^{-11}$, at which a null excludes it at $\sim 20\sigma$.

Sensitivity is Mars-dominated (MGS/MEX/MRO ranging; idealized per-dataset S/N at the prediction: Mars 21–40, Mercury/MESSENGER 1.3–4.4, Venus ~ 1.3 , Saturn/Cassini 0.6–1.0, Jupiter/Juno $\lesssim 0.1$). These are white-noise upper estimates; realistic systematics (asteroid modeling above all) degrade them by a few $\times$.

4. A provisional bound from public data

No machine-readable normal-point archive is publicly downloadable at present (the INPOP APDB service and the published Cassini normal points were not retrievable at run time), but the published **secular perihelion and node advances** (Hees et al. 2015, Table I; INPOP15a updates from Fienga et al. 2016) permit the fixed-direction fit directly. Fitting the single amplitude A to the ten secular observables (Earth–Moon barycenter excluded on ω -definition grounds; Venus carries a $1/e$ enhancement making it competitive despite weaker ranging):

$$A = (-2.5 \pm 5.9) \times 10^{-10} \text{ (INPOP10a set)} \cdot A = (-0.9 \pm 6.4) \times 10^{-10} \text{ (INPOP15a-updated set)} \text{ 95\% bound } |A| < 1.2 \times 10^{-9} \cdot \text{prediction } -8.7 \times 10^{-10} \text{ sits at } \mathbf{1.4\sigma} \text{ (}\mathbf{1.2\sigma}\text{) with matching sign.}$$

Direction-fixing alone thus already improves on the published marginal $\bar{s}^{\{TX\}}$ bound by roughly an order of magnitude (and on the combined-analysis bound $\sim 2\times$) using nothing but public tables. We state plainly what this is not: it is not a detection (1.4σ), the sign agreement with the published central value is at present a coincidence-level observation, and the secular-table fit inherits whatever systematics the underlying ephemeris solutions carry. It is a floor-demonstration: the dedicated fit of §3 on archival ranging decides the question.

5. Scope, credit, and the pre-registration clause

The SME formalism and the ephemeris methodology are Bailey & Kostelecký (2006) and Hees et al. (2015); the Planck dipole fixes the apex; the framework supplies only the *hypothesis* that fixes the direction and the target amplitude, and it is a one-parameter reframing of the MOND scale whose earlier over-claims were publicly retracted (2026-06-23) and are not reasserted. The prediction, recipe, fork-scoping, and kill condition above were committed to the public repository (github.com/carlzimmerman/zimmerman-formula, PREREGISTERED_EMPIRICAL_FLEET_2026-07.md) before any dedicated ephemeris fit has been performed. If the dedicated fit returns a null at the stated sensitivity, this realization’s sharpest prediction is dead, and we will say so in those words.

References

- Q. G. Bailey & V. A. Kostelecký, *Signals for Lorentz violation in post-Newtonian gravity*, Phys. Rev. D 74 (2006) 045001 (gr-qc/0603030).
- A. Hees et al., *Testing Lorentz symmetry with planetary orbital dynamics*, Phys. Rev. D 92 (2015) 064049 (arXiv:1508.03478).
- A. Fienga et al., *Constraints on SME coefficients from INPOP planetary ephemerides* (2016) (arXiv:1601.00947).
- A. Hees et al., *Tests of Lorentz symmetry in the gravitational sector*, Universe 2 (2016) 30 (arXiv:1610.04682).
- M. Milgrom, *The Modified Dynamics as a Vacuum Effect*, Phys. Lett. A 253 (1999) 273 (astro-ph/9805346).
- Planck Collaboration, dipole $v = 369.82$ km/s toward $(l,b) = (264.02^\circ, 48.25^\circ)$.

Reproducible: reviews/stx_target.py, reviews/stx_inpop_recipe.py (both committed, exit 0). Companion pre-registration: PREREGISTERED_EMPIRICAL_FLEET_2026-07.md. Prediction DOI lineage: 10.5281/zenodo.20978308.